

Revealing Hidden Liabilities: Environmental Activism and Capital Structure

Abstract

This study identifies information revelation as a distinct channel through which environmental activism shapes corporate capital structure. Unlike real economic shocks, NGO campaigns expose pre-existing environmental liabilities that creditors had previously underpriced. Using 1,479 campaigns targeting U.S. firms (2009–2023), we find activism reduces book leverage by 3.7% and market leverage by 1.4%. Exploiting state Anti-SLAPP laws as an instrument, we establish causality. The deleveraging effect strengthens when campaigns coincide with litigation or protests, underscoring the roles of information credibility and salience. Activism also increases exposed environmental incidents, triggers credit downgrades (0.12 points), and raises loan spreads by 14–16 basis points. We contribute by positioning NGOs as adversarial information intermediaries that mitigate information asymmetry and reshape corporate financial policy.

Keywords: NGO activism; Environmental liability; Information revelation; Information asymmetry; Capital structure

JEL Classification: G32; Q51; M14; G34; D82

1. Introduction

Corporate capital structure theory has long emphasized how information asymmetry shapes debt capacity and financing choices (Myers & Majluf, 1984; Diamond, 1984). Yet, a critical class of risks remains systematically underpriced: latent environmental liabilities. Unlike operational risks or financial distress, environmental exposures are often technically complex, temporally distant, and strategically obscured by firms (Konar & Cohen, 2001; Lyon & Maxwell, 2011). This opacity creates a puzzle: why do environmental risks, frequently concealed or discounted by creditors, suddenly trigger substantial corporate deleveraging? We address this question by examining how environmental NGO activism serves as an adversarial information intermediary that surfaces hidden liabilities and forces their incorporation into capital structure decisions.

We argue that NGO activism affects corporate leverage primarily through an information revelation mechanism rather than through direct economic shocks. While pollution, climate risk exposure, and regulatory mandates represent real economic changes that alter cash flow distributions (Sautner et al., 2023), NGO activism operates differently: it uncovers and publicizes pre-existing environmental problems that creditors had previously underpriced or ignored. The liabilities themselves do not emerge anew; rather, activism transforms obscured risks into salient, credible signals. This revelation triggers information-induced financing frictions. Creditors reassess default risk, demand higher risk premiums, and tighten credit terms, while firms proactively deleverage to preserve financial flexibility, avoid covenant breaches, and signal creditworthiness. The net effect is a measurable reduction in both book and market leverage.

Our framework builds on information asymmetry and capital structure theories, positioning NGOs as a distinct class of external monitors that complement—and sometimes substitute for—traditional disclosure regimes. Information asymmetry is particularly severe for environmental

liabilities due to four factors: (1) technical complexity requiring specialized scientific expertise, (2) temporal uncertainty in liability manifestation, (3) strategic nondisclosure driven by cleanup and litigation costs, and (4) greenwashing that obscures true risk profiles (Clarkson et al., 2008; Marquis et al., 2016). While regulatory programs like the EPA's Toxics Release Inventory provide baseline transparency, their scope is limited and enforcement is inconsistent (Khanna et al., 1998). Voluntary disclosures face credibility challenges when firms have incentives to withhold negative information (Verrecchia, 1983; Dye, 1985). Environmental NGOs bridge this gap through adversarial monitoring. Unlike traditional information intermediaries (analysts, auditors, rating agencies) who verify disclosed fundamentals, NGOs actively investigate and publicize concealed problems. They reduce asymmetry through three channels: forcing disclosure by making corporate silence costly, providing independent verification through technical expertise, and generating salience through media amplification that overcomes creditors' attention constraints (Hirshleifer & Teoh, 2003).

Using comprehensive data on 1,479 NGO environmental campaigns targeting 572 U.S. publicly traded firms from 2009–2023, we document that activism reduces book leverage by 3.7% and market leverage by 1.4%. Dynamic difference-in-differences analyses reveal that these effects emerge gradually, peaking one year after activism begins, consistent with the time required for information to disseminate and be priced by capital markets. To address endogeneity concerns, we exploit variation in state-level Anti-SLAPP laws that protect public participation speech, providing a quasi-exogenous source of variation in activism intensity. The results are robust to modern staggered adoption estimators (Sun & Abraham, 2021; Borusyak et al., 2022; de Chaisemartin & D'Haultfœuille, 2020), alternative event windows, and net leverage specifications that confirm genuine deleveraging rather than cash accumulation.

We provide three lines of evidence supporting the information revelation channel. First, the deleveraging effect is significantly amplified when activism coincides with environmental litigation or protests. Litigation provides court-verified credibility that validates activist claims, while protests generate high-visibility salience that forces creditor attention. Second, we trace a direct two-stage pathway: activism increases the number of publicly exposed environmental controversies, and these newly surfaced incidents subsequently drive leverage reductions. This pattern confirms that activism operates by revealing pre-existing problems rather than creating new environmental risks. Third, we document direct creditor responses: targeted firms experience credit rating downgrades of 0.12 points and loan spread increases of approximately 14–16 basis points, demonstrating that lenders price the newly revealed environmental risk and adjust financing terms accordingly.

This study makes three contributions to the literature. First, we identify information revelation as a distinct mechanism through which environmental factors influence capital structure. While recent studies document how pollution (Chang et al., 2018), climate risk (Li & Zhang, 2023), and adaptation mandates (Kovacs et al., 2025) affect leverage through direct economic channels, we show that adversarial external monitoring can unlock the capital structure implications of latent environmental liabilities through information diffusion, independent of real shocks. Second, we extend the NGO activism literature by documenting a financial policy channel. Prior research emphasizes operational and disclosure responses (Eesley & Lenox, 2006; King, 2008); we demonstrate that activism fundamentally reshapes corporate financing by altering creditor risk assessments and prompting strategic rebalancing. Third, we contribute to the literature on information intermediaries in capital markets. By positioning NGOs as adversarial, mission-driven monitors who uncover concealed risks rather than verify disclosed fundamentals, we highlight how

non-traditional stakeholders can serve as de facto information catalysts that complement formal disclosure regimes and mitigate information asymmetry.

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework and hypotheses. Section 3 describes the data, variable construction, and empirical methodology. Section 4 presents the main results, mechanism tests, and robustness analyses. Section 5 concludes.

2. Theoretical Development and Hypotheses

2.1 NGO Activism as Information Revealer

Environmental NGOs have emerged as powerful external monitors that pressure firms through media campaigns, protests, and regulatory advocacy (Spar & La Mure, 2003; Yaziji & Doh, 2009). Unlike traditional governance mechanisms that operate within corporate structures, NGOs exert influence through confrontational tactics designed to impose reputational, regulatory, and financial costs on target companies. These organizations possess distinct monitoring advantages: technical staff with environmental expertise can identify problems that escape generalist analysts; long time horizons enable investigation of slowly emerging liabilities; absence of commercial relationships ensures independence; and reputational stakes in accuracy discourage false accusations. Most importantly, NGOs credibly threaten to impose costs through campaigns that damage reputation, trigger regulatory scrutiny, and mobilize stakeholders (King, 2008).

Traditional environmental regulation relies on government agencies with limited budgets, political constraints, and information asymmetries that allow firms to conceal violations (Khanna et al., 1998). Environmental NGOs fill this regulatory gap through private monitoring and public pressure. Major organizations, including Sierra Club, Greenpeace, Environmental Defense Fund,

and Natural Resources Defense Council, combine scientific analysis, media strategy, and grassroots mobilization to target both firms' environmental impacts and disclosure practices.

We ground the premise that environmental liabilities are initially *hidden* from capital markets in three complementary theoretical lenses. First, strategic obfuscation: firms actively obscure environmental risks through complex reporting, selective disclosure, or greenwashing, as the costs of cleanup, litigation, and reputational damage create strong incentives to withhold negative information (Lyon & Maxwell, 2011; Marquis et al., 2016). Second, technical complexity and temporal uncertainty: environmental liabilities often require specialized scientific expertise to assess and manifest years or decades after initial violations, making it prohibitively costly for creditors to verify or price these risks *ex ante* (Clarkson et al., 2008; Konar & Cohen, 2001). Third, limited attention: even when environmental information is technically public (e.g., in regulatory filings), creditors facing cognitive constraints and competing information demands may fail to incorporate it into lending decisions (Hirshleifer & Teoh, 2003). NGO activism cuts through these frictions by transforming obscure or suppressed information into salient, credible, and widely disseminated signals.

Activism reduces information asymmetry through three primary channels. First, it forces disclosure by making corporate silence costly; even denials or defensive responses reveal previously concealed information (Christensen et al., 2021). Second, it provides independent verification: when NGOs with technical expertise and reputational stakes publicize environmental problems, creditors view this information as more credible than voluntary corporate disclosures (Baron & Diermeier, 2007). Third, it creates salience by generating media attention that ensures creditors incorporate environmental risks into pricing and monitoring decisions, even if the information was previously available but overlooked.

When activism reveals environmental problems, creditors rationally update default risk assessments upward, recognizing that environmental liabilities create uncertain future cash outflows through litigation, penalties, remediation, and reputational damage (Karpoff et al., 2005). This triggers information-induced financing frictions: (1) creditors demand higher risk premiums and tighten covenant monitoring (Chava & Roberts, 2008; Roberts & Sufi, 2009), (2) credit rating agencies downgrade borrowers as perceived tail risk increases, and (3) agency conflicts between equity and debt holders intensify as environmental uncertainty grows (Jensen & Meckling, 1976; Myers, 1977).

These frictions operate through a dual-channel mechanism. On the *demand side*, creditors raise borrowing costs, reduce credit supply, or refuse debt rollovers. On the *supply side*, firms proactively deleverage to preserve financial flexibility, avoid covenant breaches, signal financial health, or substitute toward equity financing when debt becomes prohibitively expensive or constrained. This strategic rebalancing aligns with pecking-order and dynamic trade-off theories, in which firms adjust their capital structure to mitigate heightened information asymmetry and financing frictions (Myers & Majluf, 1984; Graham & Leary, 2011). The net result is a reduction in both book and market leverage following activism.

Hypothesis 1: *Corporate leverage declines following NGO environmental activism, driven by information revelation about pre-existing environmental liabilities.*

2.2 Information Credibility: Environmental Litigation

The effectiveness of information revelation depends critically on credibility. Creditors may discount NGO campaigns if they perceive activists as biased, non-expert, or prone to advocacy-driven exaggeration. Environmental litigation resolves this credibility problem by providing institutional verification through court proceedings that generate verified facts via discovery,

expert testimony, and judicial oversight. Unlike voluntary disclosures or activist claims, litigation-revealed information carries the imprimatur of the legal system, significantly lowering the perceived noise in the information signal (Karpoff et al., 2005).

Activism and litigation operate as complementary information channels. Litigation provides high-credibility but low-salience verification; court records and legal filings are often buried in technical databases that creditors do not routinely monitor. Activism provides high-salience but lower-credibility dissemination; media campaigns reach capital market participants but may face skepticism. When both are present, creditors receive a dual signal: environmental problems are legally actionable (as verified by courts) and materially significant (as amplified by activists). This complementarity generates a clear theoretical prediction: if activism operates through information revelation, its capital structure effects should be strongest when litigation provides independent legal verification, regardless of whether litigation preceded activism or was triggered by it.

Hypothesis 2a: *The negative effect of NGO environmental activism on corporate leverage is stronger for firms facing environmental litigation, as litigation provides credible verification that amplifies the information revelation channel.*

2.3 Information Salience: Environmental Protests

Environmental activism campaigns generate baseline information salience through press releases, reports, and stakeholder engagement, which may reach specialized audiences monitoring corporate environmental performance. However, the effectiveness of information dissemination varies significantly with tactic intensity. Environmental protests represent the peak of information salience, transforming environmental concerns from background noise into front-page news through highly visible, disruptive events that demand immediate attention.

Protests generate salience through four theoretical mechanisms. First, they signal problem severity via costly collective action; participants incur travel, time, and legal risks, credibly demonstrating that environmental issues merit serious concern (Olson, 1965). Second, protests create observable operational disruption (e.g., facility blockades, customer deterrence), providing tangible evidence of business impact (King & Soule, 2007). Third, they signal escalation risk, indicating that firms have failed to manage stakeholder relations quietly and face mounting reputational and regulatory threats (Eesley & Lenox, 2006). Fourth, the confrontational nature of protests generates sustained media coverage that amplifies information reach beyond specialized audiences to mainstream creditors and investors (McCarthy et al., 1996).

These mechanisms ensure that activism-revealed environmental information achieves sufficient salience to penetrate limited attention constraints and influence creditor decisions. If activism operates through information revelation, its capital structure effects should be strongest when protests amplify the visibility, urgency, and perceived materiality of revealed risks.

Hypothesis 2b: *The negative effect of NGO environmental activism on corporate leverage is stronger when activism involves environmental protests, as protests increase information salience and force creditor attention to revealed environmental liabilities.*

3. Empirical Method

3.1 Data and Sample

Our analysis combines data from multiple sources. Environmental activism campaigns come from SIGWATCH, a comprehensive database that tracks the actions of approximately 12,500 activist groups—covering more than 1,000 issues and monitoring 27,000 corporations worldwide—to identify firms targeted by NGOs. As of 2024, the database contains over 87,000 activist campaigns, many of which involved multiple discrete "events." For example, to pressure

fashion brands to commit to eliminating hazardous chemicals from their production practices, Greenpeace organized a three-day "social media storm" that generated more than 10,000 tweets directed at Burberry, accompanied by coordinated protests outside Burberry stores. Because individual activism campaigns can involve multiple social and environmental causes simultaneously, we manually review each campaign and identify all instances of environmental activism as those in which the NGO raised at least one major environmental issue against the company. Appendix Table A lists the top 10 NGO activists that mobilized the most campaigns, while Appendix Table B provides illustrative examples of the different environmental issues raised in these campaigns.

We obtain accounting data from Compustat, institutional ownership data from Thomson Reuters 13F Holdings, analyst coverage from I/B/E/S, shareholder activism from Audit Analytics, environmental litigation data from the Audit Analytics Corporate Legal database, and environmental protest data come from RavenPack Edge database, environmental incidents data from RepRisk database, credit ratings data from Compustat Ratings, and loan pricing data from DealScan database. After merging these databases, we require non-missing values for all variables. The final sample comprises 64,294 firm-year observations representing over 8,532 unique firms during the 2009-2023 period. Our sample begins in 2009, which is one year before the first NGO environmental activism in 2010, and ends in 2024 one year following the last activism in 2023, thereby ensuring pre- and post-treatment periods for valid difference-in-differences estimation.

In total, we identify 1,479 NGO environmental campaigns directed at publicly traded U.S. firms during 2009-2023, targeting 572 unique firms across industries. Table 1 presents the distribution of environmental activism campaigns and targeted firm-years by Fama French 30

industry classification. Notably, firms in the Machinery, Food, Electrical Equipment, and Coal industries are most frequently targeted by activist NGOs.

[Insert Table 1 here]

3.2 Measurement

3.2.1 NGO Environmental Activism

We construct an indicator variable *NGO environmental activism* that equals 1 for firm-years within one year following an NGO environmental campaign targeting the firm, and zero otherwise. This one-year window captures the period during which information revelation effects are most pronounced and when firms are most likely to respond to NGO pressure through capital structure adjustments. The choice of a one-year window balances two considerations: it is sufficient to allow information to disseminate through capital markets and influence creditor decisions, yet reasonable to minimize contamination from confounding events.

To ensure robustness and examine the persistence of activism effects, we also construct alternative measures using two-year and three-year event windows. These alternative specifications allow me to test whether the leverage effects are temporary or persistent, and whether they strengthen or attenuate over longer horizons. For the dynamic analyses presented in Section 4.1, we additionally create event-time indicator variables (*NGO environmental activism*₋₂, *NGO environmental activism*₋₁, *NGO environmental activism*₀, *NGO environmental activism*₊₁) that equal one in the second year before, first year before, year of, and first year after activism, respectively. These dynamic indicators enable me to examine pre-trends and trace out the temporal evolution of the activism effect.

3.2.2 Financial Leverage

Following standard practice in the capital structure literature (Rajan & Zingales, 1995; Frank & Goyal, 2009; Welch, 2004), we construct two primary leverage measures. *Book leverage* equals total debt (the sum of long-term debt and short-term debt) divided by book value of total assets. *Market leverage* equals total debt divided by the sum of total debt and market value of equity. Book leverage reflects the accounting perspective and is less volatile, while market leverage incorporates market expectations about future performance and risk. Both measures are widely used in capital structure research, and examining both provides a more complete picture of firms' leverage decisions.

For robustness tests, we also construct net leverage measures that account for cash holdings. *Net book leverage* equals net debt (total debt minus cash and short-term investments) divided by net assets (total assets minus cash and short-term investments), while *Net market leverage* equals net debt divided by the sum of net debt and market value of equity. These net measures address the concern that firms might respond to activism by accumulating cash rather than genuinely deleveraging, providing a more conservative test of whether activism causes true reductions in financial leverage.

3.2.3 Other variables

To conduct the cross-sectional tests, we construct moderating variables of *Environmental litigation* and *Environmental protests*. We obtain data from the Audit Analytics Corporate Legal database to measure the *Environmental litigation* variable, which is an indicator variable that equals one if a firm is the defendant of an environmental lawsuit in a year. We obtain data on environmental protests from the RavenPack Edge database to measure *Environmental protests*, which is the number of environmental protests the firm faces in a year. To conduct the channel tests, we construct three variables. First, we obtain publicly exposed environmental incidents data

from the RepRisk database to measure *Exposed environmental incidents*, which is the number of environmental controversies and allegations. Second, we obtain credit ratings data from the Compustat S&P Rating database to measure *Credit rating*, which is the numerical rank of S&P long-term issuer credit ratings, ranging from 1 (lowest credit quality, default) to 20 (highest credit quality, AAA). Higher values indicate better creditworthiness and lower default risk. Third, we obtain loan pricing data from the DealScan database to measure *Loan spread*, which is the all-in spread drawn, measured in basis points. This represents the amount the borrower pays in basis points over LIBOR (or the applicable base rate) for each dollar drawn down from the loan facility.

We control for firm *Size*, measured as the natural logarithm of total assets, because larger firms maintain higher leverage due to lower default risk and greater access to capital markets (Rajan & Zingales, 1995; Titman & Wessels, 1988). We also control for *ROA* (equal to operating income over total assets) and *Tobin's q* (measured as the market value of assets over the book value of assets) because better performing firms may have more leverage. Next, we control for a firm's financial conditions by including *Cash*, measured as the total cash balance over total assets because cash holdings substitute for debt capacity and provide financial flexibility (Opler et al., 1999). We control for *Dividend* (common dividends over total assets), *Tangibility* (tangible assets over total assets), *Capital expenditures* (capital expenditures over lagged total assets), *R&D* (research and development expenses over total assets), and *M&A* (the number of asset acquisitions) because firms may increase leverage to fund these capital policies. Lastly, we control for three governance-related variables—*Institutional ownership* as the percentage of shares owned by institutional investors, *Shareholder activism*, coded as 1 in years in which a Schedule 13D was filed with the SEC, and 0 otherwise, and *Analyst coverage* as the natural logarithm of 1 plus the number of analyst forecasts issued for a firm. We add these variables because institutional investors, activist

shareholders, and financial analysts can influence corporate decisions via governance and information channels. Appendix C provides detailed variable definitions and data sources.

Table 2 presents summary statistics. Average book leverage is 0.430 with substantial cross-sectional variation (standard deviation 1.182). Market leverage averages 0.221 (standard deviation 0.244). Approximately 3.6% of firm-years' experience in NGO environmental activism. Environmental litigation occurs in 1.8% of firm-years, while environmental protests occur in 9.0% of sample firm-years.

[Insert Table 2 here]

3.3 Model Specification

The primary specification employs staggered difference-in-differences regressions to test how corporate leverage responds to NGO environmental activism that occurs in different years, spanning from the pre-activism to the post-activism periods. Specifically, we estimate the specification:

$$\begin{aligned} Leverage_{i,j,t} = & \alpha + \beta \times NGO\ environmental\ activism_{i,t} + \gamma \times Controls_{i,t} + \delta_i \\ & + \lambda_{j,t} + \varepsilon_{i,j,t} \quad (1) \end{aligned}$$

where $Leverage_{i,j,t}$ represents book or market leverage for firm i in year t , $NGO\ environmental\ activism_{i,t}$ equals 1 for the firm-year observations within the following 2 year after NGO environmental activism and zero otherwise. β measures the difference of the average change in $Leverage_{i,j,t}$ after the activism for target firms and control firms who do not experience the activism. We choose a two-year window and exclude observations beyond 2 year for treatment firms to minimize the confounding effects. In addition, we report the robustness results using 1- and 3-year windows. $Controls_{i,t}$ includes the firm-level controls described above, δ_i is firm fixed effects controlling for time-invariant firm heterogeneity, $\lambda_{j,t}$ is industry-year fixed

effects (based on 2-digit SIC codes) controlling for industry-specific time trends, and $\varepsilon_{i,j,t}$ is error terms. Standard errors are clustered at the firm level. The firm fixed effects specification identifies effects from within-firm variation over time, controlling for all time-invariant firm characteristics including baseline environmental risk, industry membership, and managerial style. Industry-year fixed effects account for common shocks affecting all firms in an industry (e.g., regulatory changes, commodity price movements) that might correlate with both activism and leverage.

For cross-sectional analyses testing Hypotheses 2a and 2b, we estimate:

$$\begin{aligned} \text{Leverage}_{i,j,t} = & \alpha + \beta_1 \times \text{NGO environmental activism}_{i,t} \\ & + \beta_2 \times \text{NGO environmental activism}_{i,t} \times \text{Moderator}_{i,t} \\ & + \beta_3 \times \text{Moderator}_{i,t} + \gamma \times \text{Controls}_{i,t} + \delta_i + \lambda_{j,t} + \varepsilon_{i,j,t} \quad (2) \end{aligned}$$

where $\text{Moderator}_{i,t}$ represents environmental litigation or environmental protests. The coefficient β_2 captures how the moderator amplifies or attenuates the activism effect.

4. Results

4.1 Baseline Results

Table 3 presents our main results. Column (1) shows a univariate specification where the coefficient on *NGO environmental activism* is -0.026 ($p < 0.05$), indicating book leverage decreases by approximately 2.6% following activism. Column (2) adds controls, yielding a coefficient of -0.037 ($p < 0.01$), an 8.6% decrease relative to the mean book leverage of 0.430.

[Insert Table 3 here]

Column (3) examines the dynamic effects using event-time dummies. Pre-activism coefficients are small and statistically insignificant (*activism*₋₂: 0.003, $p > 0.10$; *activism*₋₁: -0.017, $p > 0.10$), supporting causal interpretation by showing no pre-existing trends. The activism effect emerges in the year of activism (*activism*₀: -0.032, $p < 0.10$) and strengthens in the following year

($activism_{+1}$: -0.055, $p < 0.01$), consistent with information dissemination and gradual incorporation into lending decisions.

Columns (4)-(6) show market leverage results. Column (4) presents a univariate specification with a coefficient of -0.014 ($p < 0.05$). Column (5) shows that with controls, market leverage decreases by 1.4 percentage points ($p < 0.01$), a 6.3% reduction relative to the mean of 0.221. The dynamic specification in Column (6) again shows no pre-trends ($activism_{-2}$: -0.012, $p > 0.10$; $activism_{-1}$: -0.012, $p > 0.10$) and strengthening effects over time ($activism_0$: -0.016, $p < 0.10$; $activism_{+1}$: -0.027, $p < 0.01$).

The control variables show expected signs. Larger firms have lower book leverage but slightly higher market leverage. Cash holdings strongly reduce leverage, consistent with pecking order theory. Tangibility increases leverage, consistent with collateral value. Having established the main effect, we now turn to cross-sectional variation that tests the information revelation mechanism.

4.2 Identification: Heckman Treatment Effect Model

A central identification concern is selection bias: NGOs may strategically target firms with specific characteristics correlated with leverage decisions, potentially confounding the causal interpretation of our findings. For instance, activists may preferentially target larger, more visible firms or those with particularly poor environmental records. If these same characteristics independently predict leverage changes, the observed association between activism and deleveraging could reflect selection rather than causation. To address this endogeneity concern, we employ a Heckman treatment effect model using state-level anti-SLAPP (Strategic Lawsuit Against Public Participation) laws as an instrumental variable for NGO environmental activism.

Anti-SLAPP laws provide legal protections that dismiss meritless lawsuits designed to silence public participation on matters of public concern. These statutes allow defendants to file special motions to strike SLAPP suits at early litigation stages, shift attorney fee burdens to plaintiffs filing frivolous claims, and provide immunity for good-faith public advocacy (Wu et al., 2025). By reducing the legal risks and costs of advocacy, anti-SLAPP laws lower barriers to NGO environmental activism. The Heckman treatment effect model addresses potential selection bias by explicitly modeling both the treatment assignment (activism) and the outcome (leverage) equations (Heckman, 1979; Lennox et al., 2012). As Lennox et al. (2012) emphasize, the validity of this approach critically depends on identifying appropriate exclusion restrictions—variables that affect treatment assignment but not the outcome except through treatment. The anti-SLAPP laws serve this purpose in our setting.

The validity of anti-SLAPP laws as an instrument requires satisfying two conditions. First, the *relevance condition* requires that anti-SLAPP laws predict NGO environmental activism. The legal protections provided by these statutes directly reduce the costs and risks of activism by making it more difficult for corporations to retaliate through defamation suits or other legal harassment. When activists face lower litigation risk, they can more aggressively investigate and publicize corporate environmental problems. This institutional mechanism generates a strong first-stage relationship between anti-SLAPP law adoption and activism incidence.

Second, the *exclusion restriction* requires that anti-SLAPP laws affect corporate leverage only through their effect on activism, not through other channels. This assumption is plausible for several reasons. Anti-SLAPP laws specifically protect public participation in environmental and policy debates rather than regulating corporate environmental practices directly. The laws impose no compliance costs on corporations, mandate no operational changes, and create no

environmental liabilities beyond those revealed through activism. Unlike environmental regulations that directly constrain production or impose cleanup obligations, anti-SLAPP laws operate exclusively by facilitating third-party monitoring and information revelation. The laws do not alter firms' actual environmental risk exposure, financial constraints, or investment opportunities. Their sole mechanism affecting leverage operates through enabling NGO activism that reveals environmental information to creditors.

Panel A of Table 4 presents first-stage probit regression results. The coefficient on Anti-SLAPP is 0.272 ($p < 0.01$), indicating that firms headquartered in states with anti-SLAPP protections face substantially higher activism likelihood. The F-statistic of 29.680 exceeds conventional thresholds (e.g., 10) for instrument strength, confirming that anti-SLAPP laws strongly predict activism and satisfy the relevance condition.

[Insert Table 4 here]

Panel B presents second-stage results estimating the effect of instrumented activism on leverage. Book leverage decreases by 3.8% ($p < 0.01$) and market leverage decreases by 2.0% ($p < 0.01$) following activism. These magnitudes slightly exceed the baseline OLS estimates in Table 3, consistent with two possibilities: measurement error in the activism variable attenuating OLS estimates, or unobserved firm characteristics that make activism less likely (e.g., proactive environmental management) also independently reducing leverage, causing OLS to understate true effects. Importantly, the inverse Mills ratio—capturing selection into treatment—is statistically insignificant in both specifications, suggesting that selection bias does not substantively confound the baseline estimates.

The Heckman treatment effect model results support causal interpretation of the activism-leverage relationship. By exploiting quasi-random variation in activism likelihood induced by

state-level legal protections, the instrumental variables approach addresses selection concerns and confirms that NGO environmental activism causally reduces corporate leverage. The consistency between OLS and IV estimates, combined with the strong first-stage relationship and plausible exclusion restriction, strengthens confidence that information revelation about environmental liabilities drives the observed deleveraging.

4.3 The Moderating Effects with Environmental Litigation and Protests

4.3.1 Environmental Litigation for Information Credibility

Columns (1) and (2) of Table 5 examine whether environmental litigation amplifies the activism effect. The interaction term $\text{NGO activism} \times \text{Environmental litigation}$ is -0.059 ($p < 0.05$) for book leverage and -0.035 ($p < 0.05$) for market leverage. These results indicate that firms facing both activism and lawsuits reduce book leverage by an additional 5.9% beyond the baseline activism effect of -0.029 —more than doubling the impact. For market leverage, the additional reduction is 3.5% beyond the baseline effect of -0.010 .

[Insert Table 5 here]

These results support Hypothesis 2a and confirm that environmental litigation amplifies the activism effect by increasing information credibility about environmental liabilities. Litigation provides credible, court-verified information about environmental problems through discovery processes, expert testimony, and judicial oversight. When combined with activism's broad dissemination through media campaigns and public advocacy, the dual signal creates particularly powerful information revelation. Creditors viewing both litigation and activism rationally infer that environmental problems are both legally actionable (litigation occurred) and systemically significant (activism targets the firm), triggering especially strong deleveraging responses.

4.3.2 Environmental Protests for Information Salience

Columns (3) and (4) of Table 5 examine whether environmental protests amplify the activism effect. The interaction term $\text{NGO activism} \times \text{Environmental protests}$ is -0.001 ($p < 0.05$) for book leverage and -0.001 ($p < 0.01$) for market leverage. Given that the environmental protests variable represents the count of protests, these coefficients indicate that each additional protest amplifies the activism effect. The baseline activism effect is -0.019 ($p < 0.05$) for book leverage and -0.011 ($p < 0.05$) for market leverage.

These results support Hypothesis 2b and confirm that environmental protests amplify the activism effect by increasing information salience of environmental liabilities. Protests demonstrate serious stakeholder concerns through costly collective action—participants incur travel costs, time commitments, and potential legal risks. This visible mobilization creates observable operational disruption (blocking facility access, deterring customers), attracts extensive media coverage due to the visual and confrontational nature, signals escalation potential to creditors who recognize that firms facing protests have failed to manage stakeholder relations quietly, and generates reputational damage extending beyond immediate participants. The cumulative effect makes environmental information particularly credible and salient to creditors, who rationally infer that problems are sufficiently severe to motivate costly collective action.

4.4 Information Revelation of NGO Activism on Environmental Liabilities

Panel A shows that NGO environmental activism increases the number of exposed environmental incidents by 0.285 ($p < 0.01$). This finding directly supports the core theoretical argument: activism reveals hidden environmental problems that already existed but were previously concealed or unreported, bringing them into public view and making them known to capital markets. Critically, these newly exposed incidents represent pre-existing environmental problems—pollution releases, regulatory violations, or hazardous waste mismanagement that had

occurred but remained hidden from creditors—rather than new environmental problems created by the activism itself.

[Insert Table 6 here]

Panel B demonstrates the consequences of this revelation. *Exposed environmental incidents* reduce book leverage by 0.3% per incident ($p < 0.05$) and market leverage by 0.4% per incident ($p < 0.01$). The effects are economically meaningful— a one standard deviation increase (1.749) of *Exposed environmental incidents* leads to a 0.5% decrease of book leverage and 0.7% decrease of market leverage, accounting for part of the total activism effect documented in Table 3. These results confirm that information revelation about previously hidden environmental liabilities serves as a channel through which activism affects capital structure.

Together, these results provide strong support for the information revelation mechanism. Activism does not merely correlate with leverage reductions—it demonstrably increases the number of publicly known environmental incidents by exposing problems that were previously concealed, and these newly exposed incidents themselves reduce leverage. This two-stage process (activism exposes previously hidden incidents, which leads to deleveraging) confirms that activism works by revealing previously hidden information rather than creating new environmental risks. The pattern is consistent with NGOs uncovering and publicizing environmental problems through investigative research, facility inspections, and analysis of regulatory filings—bringing to light incidents that occurred in the past but were not widely known to capital market participants until activism made them salient.

4.5 The Effects of NGO Activism on Credit Ratings

The information revelation mechanism predicts that NGO activism affects leverage decisions by prompting creditors to reassess and price environmental risk. If activism reveals

previously hidden environmental liabilities, credit rating agencies—which systematically evaluate default risk—should incorporate this information into their ratings. Credit downgrades following activism would provide direct evidence that the market perceives increased credit risk, validating that activism reveals material information about environmental liabilities rather than merely creating noise or temporary reputational concerns.

[Insert Table 7 here]

Table 7 examines whether NGO environmental activism affects credit ratings. The results show that NGO environmental activism reduces credit ratings by 0.120 points ($p < 0.05$), indicating that activism leads to credit downgrades. Credit ratings range from 1 (lowest quality) to 20 (highest quality, AAA), so this represents a meaningful deterioration in creditworthiness. This finding provides additional support for the information revelation mechanism—as activism exposes environmental liabilities, credit rating agencies incorporate this information into their assessments, downgrading firms with revealed environmental risks.

4.6 The Effects of NGO Activism on the Cost of Debt

Beyond credit ratings, the information revelation mechanism implies that creditors should demand compensation for newly revealed environmental risks through higher interest rates. When activism exposes environmental liabilities that creditors had previously underestimated, lenders should respond by charging risk premiums on new loans to offset the increased probability of default and expected losses. Examining loan spreads—the interest rate premium over the risk-free rate—provides a direct test of whether creditors price the environmental information revealed through activism. An increase in loan spreads following activism would demonstrate that creditors view revealed environmental liabilities as material risks requiring compensation, consistent with information revelation affecting the cost of debt.

[Insert Table 8 here]

Table 8 examines whether NGO environmental activism affects the cost of debt. The results show that NGO environmental activism increases loan spreads by 14.256 basis points ($p < 0.05$) in Column (1), 15.774 basis points ($p < 0.05$) in Column (2), and 14.493 basis points ($p < 0.05$) in Column (3). These estimates are consistent across different specifications controlling for loan type fixed effects, loan purpose fixed effects, and both. The increase in loan spreads provides direct evidence that creditors charge higher interest rates to compensate for the environmental risks revealed through activism, supporting the information revelation mechanism.

4.7 Robustness Tests

Table 9 presents extensive robustness tests addressing potential identification concerns and demonstrating the robustness of our findings. The top panel examines book leverage, while the bottom panel examines market leverage.

[Insert Table 9 here]

4.7.1 Alternative DID Estimators

Recent econometric research highlights potential biases in two-way fixed effects models with staggered treatment timing and heterogeneous treatment effects (Goodman-Bacon, 2021; Sun & Abraham, 2021; de Chaisemartin & D’Haultfœuille, 2020). We address these concerns using three alternative estimators. Column (1) implements the interaction-weighted estimator of Sun and Abraham (2021), yielding a coefficient of -0.033 ($p < 0.01$) for book leverage and a coefficient of -0.012 ($p < 0.01$). Column (2) applies the imputation-based approach of Borusyak et al. (2022), producing a coefficient of -0.044 ($p < 0.01$) for book leverage and a coefficient of -0.009 ($p < 0.05$). Column (3) uses the modified two-way fixed effects estimator of de Chaisemartin and D’Haultfœuille (2020), generating a coefficient of -0.032 ($p < 0.01$) for book leverage and a

coefficient of -0.012 ($p < 0.05$). All three methods yield estimates consistent with the baseline results, confirming that our findings are not artifacts of specification choice or econometric concerns about staggered treatment timing.

4.7.2 Alternative Event Windows

The baseline specification uses a one-year post-activism window to minimize confounding from subsequent events. Columns (4) and (5) examine whether results are sensitive to this choice. Column (4) extends the window to two years, yielding a coefficient of -0.033 ($p < 0.05$) for book leverage and a coefficient of -0.016 ($p < 0.01$) for market leverage. Column (5) extends the window to three years, producing a coefficient of -0.034 ($p < 0.05$) for book leverage and a coefficient of -0.019 ($p < 0.01$) for market leverage. The consistency across windows suggests that activism's effects are persistent rather than temporary, and that the results are not sensitive to the specific event window definition.

4.7.3 Alternative Dependent Variable

Column (6) uses net leverage measures that subtract cash holdings from debt and assets, addressing concerns that firms might simply accumulate cash rather than truly deleveraging. Net book leverage decreases by 15.9% ($p < 0.01$), and net market leverage decreases by 1.9% ($p < 0.05$). These substantially larger effects indicate that firms engage in genuine deleveraging rather than merely building cash buffers. The magnitudes suggest that some firms may be simultaneously reducing debt and accumulating cash as defensive responses to revealed environmental risks.

Collectively, these robustness tests address the primary threats to identification in this setting. The alternative DID estimators confirm that staggered treatment timing does not bias the estimates. The dynamic specifications in Table 3 show no pre-trends, ruling out anticipation effects or selection on trends. The consistency across event windows demonstrates persistence. The net

leverage results confirm genuine deleveraging rather than cash accumulation. Together, these tests support a causal interpretation of our findings: NGO environmental activism causes firms to reduce leverage through information revelation about environmental liabilities.

5. Conclusion

This study examines how environmental NGO activism influences corporate capital structure. We document that activist campaigns trigger significant reductions in both book and market leverage, operating primarily through an information revelation mechanism rather than through direct economic shocks. By surfacing pre-existing, concealed environmental liabilities, activism transforms latent risks into salient, credible signals that creditors incorporate into lending decisions, inducing firms to deleverage through a dual channel of tightened credit supply and strategic financial rebalancing.

Using a comprehensive dataset of 1,479 campaigns targeting 572 U.S. publicly traded firms from 2009 to 2023, we find that activism reduces book leverage by 3.7% and market leverage by 1.4%. These effects emerge gradually, peaking one year post-activism, consistent with information diffusion and creditor reassessment. The results are robust to modern staggered difference-in-differences estimators, alternative event windows, and net leverage specifications that confirm genuine deleveraging rather than cash accumulation.

We provide three lines of evidence supporting the information channel. First, the deleveraging effect is substantially amplified when activism coincides with environmental litigation or protests, demonstrating that legal credibility and public salience jointly strengthen the information signal. Second, activism significantly increases the number of publicly exposed environmental incidents, which in turn drive leverage reductions, tracing a clear two-stage pathway from information discovery to capital structure adjustment. Third, targeted firms

experience credit rating downgrades and loan spread increases of approximately 14–16 basis points, providing direct evidence that creditors price the newly revealed risks and adjust financing terms accordingly.

These findings contribute to several strands of literature. First, we identify information revelation as a distinct mechanism through which environmental factors shape capital structure, complementing recent work on pollution, climate risk, and regulatory mandates that operate through real economic channels. Second, we extend the NGO activism literature by documenting a financial policy response, showing that external stakeholder pressure reshapes corporate financing by altering creditor risk assessments and prompting strategic deleveraging. Third, we position NGOs as adversarial information intermediaries who uncover concealed risks rather than verify disclosed fundamentals, highlighting how non-traditional monitors can mitigate information asymmetry and complement formal disclosure regimes.

The results carry important implications for corporate governance, capital markets, and environmental policy. For creditors and investors, our findings underscore the materiality of non-financial information and the value of external monitoring in pricing latent liabilities. For firms, they highlight the financial consequences of environmental opacity and the strategic imperative of proactive risk management and transparent disclosure. For policymakers and regulators, they suggest that adversarial stakeholder monitoring can serve as a *de facto* information catalyst, improving market efficiency in domains where traditional oversight is constrained by technical complexity, limited enforcement capacity, or strategic obfuscation.

Our study has limitations that invite future research. First, while our identification strategy addresses selection bias, unobserved firm-level dynamics may still influence the timing or intensity of activism campaigns. Second, our analysis focuses on U.S. publicly traded firms; extending this

framework to private firms, emerging markets, or jurisdictions with different legal and regulatory environments would enhance generalizability. Third, future work could explore how firms' strategic responses to activism—such as enhanced voluntary disclosure, board restructuring, or sustainability investments—moderate or amplify the capital structure effects. Finally, examining whether NGO-driven information revelation has spillover effects on peer firms' financing decisions or industry-wide capital allocation would deepen our understanding of networked information diffusion in capital markets.

Ultimately, this study demonstrates that environmental activism does more than shape corporate reputation or operational practices; it fundamentally alters the information environment in which capital structure decisions are made. By revealing hidden liabilities and forcing their incorporation into creditor risk assessments, NGOs serve as critical information intermediaries that bridge the gap between corporate environmental performance and financial market outcomes. As environmental risks grow in complexity and scale, understanding how adversarial monitoring shapes corporate financing will remain essential for both theory and practice.

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Appendix A. Variable definitions

Variable	Definition	Data source
NGO environmental activism	An indicator variable that equals 1 for the firm-year observations within the following 2 year after NGO environmental activism and zero otherwise.	SIGWATCH
Market leverage	The ratio of long-term and short-term debt to the sum of long-term debt, short-term debt, and market value of equity. Formula: $(DLTT + DLC) / [(DLTT + DLC) + (PRCC_F \times CSHO)]$	Compustat
Book leverage	The ratio of long-term and short-term debt to total assets. Formula: $(DLTT + DLC) / AT$	Compustat
Net book leverage	The ratio of net debt (long-term debt plus short-term debt minus cash) to net assets (total assets minus cash). Formula: $[(DLTT + DLC) - CHE] / (AT - CHE)$	Compustat
Net market leverage	The ratio of net debt (long-term debt plus short-term debt minus cash) to the sum of net debt and market value of equity. Formula: $[(DLTT + DLC) - CHE] / [(DLTT + DLC) - CHE + (PRCC_F \times CSHO)]$	Compustat
Long-term market leverage	The ratio of long-term debt to the sum of long-term debt and market value of equity. Formula: $DLTT / [DLTT + (PRCC_F \times CSHO)]$	Compustat
Short-term market leverage	The ratio of short-term debt to the sum of short-term debt and market value of equity. Formula: $DLC / [DLC + (PRCC_F \times CSHO)]$	Compustat
Net debt issuance	The change in total debt (long-term plus short-term) scaled by market value of equity. Formula: $(DLTT + DLC) - (Lag\ DLTT + Lag\ DLC) / (PRCC_F \times CSHO)$	Compustat
Net equity issuance	The change in book equity not attributable to retained earnings, scaled by market value of equity. Formula: $[(CEQ - Lag\ CEQ) - (NI - DVC - DVP)] / (PRCC_F \times CSHO)$	Compustat
Size	The natural logarithm of book value of total assets.	Compustat
ROA	The ratio of operating income to lagged total assets.	Compustat
Tobin's q	The ratio of market value of assets (book value of total assets minus book value of equity minus deferred taxes plus market value of equity) to book value of total assets.	Compustat
Cash	The ratio of cash and short-term investment to total assets.	Compustat
Dividend	The ratio of common dividends to lagged total assets.	Compustat
Tangibility	The ratio of plant, property, and equipment over total assets.	Compustat
Capital expenditures	The ratio of capital expenditures to lagged total assets.	Compustat
R&D	The ratio of research and development expenses to lagged total assets.	Compustat

M&A	An indicator variable that equals 1 if a firm acquires any assets.	SDC
Institutional ownership	The percentage of shares outstanding owned by institutional investors.	TR 13F
Shareholder activism	An indicator variable that equals 1 if a firm has at least one 13D Filing in a year.	Audit Analytics
Analyst coverage	The average number of earnings forecasts by analysts.	IBES
Anti-SLAPP law	An indicator variable that equals one if a firm is headquartered in a state that adopted an Anti-Strategic Lawsuits Against Public Participation in a year and zero otherwise.	Lee et al. (2025)
State environmental NGOs	The number of environmental activist NGOs in a headquarter state in a year.	National Center for Charitable Statistics
Environmental Refinitiv overall rating	Refinitiv's composite rating of the firm's overall environmental performance.	LSEG Refinitiv
Environmental emission rating	Refinitiv's rating of emissions management.	LSEG Refinitiv
Environmental resource use rating	Refinitiv's rating of resource use efficiency.	LSEG Refinitiv
Environmental innovation rating	Refinitiv's rating of environmental technology innovation.	LSEG Refinitiv
Exposed environmental liabilities	The number of environmental controversies and allegations	RepRisk
Credit ratings	The numerical rank of S&P long-term issuer credit ratings, ranging from 1 (lowest credit quality, default) to 20 (highest credit quality, AAA). Higher values indicate better creditworthiness and lower default risk. The variable captures the borrower's credit quality at loan origination based on S&P's assessment of the issuer's ability to meet long-term financial obligations.	Compustat
Loan spread	The all-in spread drawn, measured in basis points, represents the amount the borrower pays in basis points over LIBOR (or the applicable base rate) for each dollar drawn down from the loan facility. This spread reflects the credit risk premium and represents the lender's compensation for extending credit to the borrower.	DealScan

Table 1. NGO environmental activism campaigns by Fama French 30 Industry

Fama French 30 Industry	Campaigns	Target firm-years
Aircraft	30	21
Apparel	32	22
Automobiles and Trucks	14	7
Beer & Liquor	19	4
Chemicals	9	6
Coal	86	54
Communication	35	20
Construction	17	11
Construction Materials	9	7
Consumer Goods	28	18
Defense	4	3
Electrical Equipment	311	102
Entertainment	1	1
Fabricated Products	30	10
Food	293	45
Healthcare	41	21
Machinery	289	98
Medical Equipment	1	1
Non-Metallic and Industrial Metal Mining	34	23
Petroleum and Natural Gas	26	19
Pharmaceutical Products	5	4
Precious Metals	53	18
Printing and Publishing	31	19
Recreation	15	5
Rubber and Plastic Products	7	6
Shipbuilding, Railroad Equipment	26	12
Steel Works Etc	30	12
Textiles	1	1
Tobacco Products	2	2
Total	1479	572

Note: This table presents the number of NGO environmental activism campaigns targeted firm-years by industry.

Table 2. Summary statistics

Variable	N	Mean	SD.	P25	Median	P75
Market leverage	40,974	0.219	0.228	0.023	0.148	0.344
Book leverage	40,974	0.261	0.244	0.047	0.221	0.392
Long-term market leverage	40,974	0.192	0.216	0.009	0.119	0.304
Short-term market leverage	40,974	0.051	0.110	0.000	0.009	0.044
Net debt issuance	40,974	0.021	0.114	-0.014	0.000	0.045
Net equity issuance	40,974	0.142	0.368	-0.010	0.012	0.100
Size	40,974	6.767	2.133	5.293	6.775	8.221
ROA	40,974	-0.029	0.398	-0.041	0.093	0.156
Tobin's q	40,974	2.347	2.005	1.177	1.654	2.684
Cash	40,974	0.260	0.274	0.052	0.146	0.390
Dividend	40,974	0.014	0.031	0.000	0.000	0.015
Tangibility	40,974	0.249	0.249	0.058	0.148	0.374
Capital expenditures	40,974	0.050	0.068	0.012	0.028	0.060
R&D	40,974	0.103	0.205	0.000	0.011	0.110
M&A	40,974	0.298	0.512	0.000	0.000	0.693
Institutional ownership	40,974	0.562	0.339	0.249	0.639	0.873
Shareholder activism	40,974	0.083	0.276	0.000	0.000	0.000
Analyst coverage	40,974	1.612	0.620	1.099	1.582	2.076
Anti-SLAPP	38,073	0.615	0.487	0.000	1.000	1.000
State environmental NGOs	38,073	147.600	174.600	0.000	99.000	186.000
Environmental overall rating	11,127	0.400	0.302	0.135	0.268	0.684
Environmental emission rating	11,127	0.259	0.313	0.000	0.099	0.489
Environmental resource use rating	11,127	0.238	0.294	0.000	0.096	0.437
Environmental innovation rating	11,127	0.169	0.266	0.000	0.000	0.344
Exposed environmental incidents	40,974	0.187	0.812	0.000	0.000	0.000
Credit rating	6,490	10.233	3.227	8.000	10.000	13.000
Loan spread	7,690	223.450	119.831	145.000	200.000	275.000

Note: All continuous variables are winsorized at 1% and 99%.

Table 3. NGO environmental activism and capital structure

Dependent variable	Market leverage			Book leverage	Net market leverage	Net book leverage
	(1)	(2)	(3)	(4)	(5)	(6)
NGO environmental activism	-0.020*** (0.007)	-0.018*** (0.007)		-0.013** (0.006)	-0.025*** (0.010)	-0.184*** (0.049)
NGO environmental activism ₋₂			-0.002 (0.008)			
NGO environmental activism ₋₁			-0.010 (0.009)			
NGO environmental activism ₀			-0.015 (0.009)			
NGO environmental activism ₊₁			-0.023** (0.010)			
NGO environmental activism ₊₂			-0.029*** (0.011)			
Size		0.020*** (0.003)	0.020*** (0.003)	0.001 (0.004)	0.011 (0.012)	-0.616*** (0.075)
ROA		-0.061*** (0.006)	-0.061*** (0.006)	-0.049*** (0.010)	-0.030 (0.028)	0.811*** (0.180)
Tobin's q		-0.016*** (0.001)	-0.016*** (0.001)	0.003** (0.001)	0.033*** (0.003)	0.090*** (0.022)
Cash		-0.172*** (0.010)	-0.171*** (0.010)	-0.221*** (0.016)	-0.759*** (0.056)	-9.694*** (0.355)
Dividend		-0.217*** (0.038)	-0.217*** (0.038)	-0.041 (0.060)	-0.363* (0.206)	-1.848** (0.832)
Tangibility		0.218*** (0.021)	0.218*** (0.021)	0.192*** (0.027)	0.261*** (0.062)	0.150 (0.306)
Capital expenditures		-0.317***	-0.317***	-0.235***	-0.254***	1.057***

		(0.023)	(0.023)	(0.027)	(0.067)	(0.341)
R&D		-0.069***	-0.069***	-0.084***	-0.157**	-2.298***
		(0.011)	(0.011)	(0.019)	(0.066)	(0.430)
M&A		-0.013***	-0.013***	-0.003	-0.010**	0.023
		(0.002)	(0.002)	(0.002)	(0.004)	(0.016)
Institutional ownership		-0.075***	-0.075***	-0.035***	-0.063**	0.353**
		(0.008)	(0.008)	(0.011)	(0.032)	(0.149)
Shareholder activism		0.007**	0.007**	-0.003	0.003	-0.127**
		(0.003)	(0.003)	(0.004)	(0.014)	(0.056)
Analyst coverage		-0.034***	-0.034***	0.008	-0.050***	0.674***
		(0.005)	(0.005)	(0.006)	(0.017)	(0.090)
Firm FE	Y	Y	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y	Y	Y
Adjusted R ²	0.761	0.791	0.791	0.739	0.349	0.740
Observations	40,974	40,974	40,974	40,974	40,974	40,974

Note: This table presents the results of estimating the effect of NGO environmental activism on financial leverage. The sample consists of firms from the intersection of SIGWATCH data and Compustat data for the period 2009-2023. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Market leverage* is the ratio of long-term and short-term debt to the sum of long-term debt, short-term debt, and market value of equity. *Book leverage* is the ratio of long-term and short-term debt to total assets. Variable definitions are in Appendix C. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 4. Capital structure by horizon and class

Panel A. Long- and short-term debt		
Dependent variable	Long-term market leverage (1)	Short-term market leverage (2)
NGO environmental activism	-0.017*** (0.006)	-0.008** (0.004)
Controls	Y	Y
Firm FE	Y	Y
Industry $\times$ Year FE	Y	Y
Adjusted R ²	0.772	0.608
Observations	40,974	40,974
Panel B. Debt and equity		
Dependent variable	Net debt issuance (1)	Net equity issuance (2)
NGO environmental activism	-0.009** (0.004)	0.019*** (0.005)
Controls	Y	Y
Firm FE	Y	Y
Industry $\times$ Year FE	Y	Y
Adjusted R ²	0.163	0.722
Observations	40,974	40,974

Note: This table presents the results of estimating the effect of NGO environmental activism on capital horizon and class. The sample consists of target and non-target firms from the intersection of SIGWATCH data and Compustat data for the period 2009-2023. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Long-term market leverage* is the ratio of long-term debt to the sum of long-term debt and market value of equity. *Short-term market leverage* is the ratio of short-term debt to the sum of short-term debt and market value of equity. *Net debt issuance* is the change in total debt (long-term plus short-term) scaled by market value of equity. *Net equity issuance* is the change in book equity not attributable to retained earnings, scaled by market value of equity. Variable definitions are in Appendix A. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 5. Heckman treatment effect model

Panel A. The first stage regression	
Dependent variable	NGO environmental activism (1)
Anti-SLAPP	0.170*** (0.059)
State Environmental NGOs	0.000** (0.000)
Chi-square statistic	19.470
P-value	0.000
Model	Probit
Controls	Y
Industry FE	Y
Year FE	Y
Pseudo R ²	0.287
Observations	38,073
Panel B. The second stage regression	
Dependent variable	Market leverage (1)
NGO environmental activism	-0.021*** (0.008)
Inverse mills ratio	0.003 (0.002)
Controls	Y
Firm FE	Y
Industry × Year FE	Y
Adjusted R ²	0.790
Observations	38,073

Note: This table presents the results of estimating the two-stage Heckman treatment effect model using the Anti-SLAPP law and local environmental NGOs as the instruments for NGO environmental activism on financial leverage. The sample consists of target and non-target firms from the intersection of SIGWATCH data and Compustat data for the period 2009-2023. *Anti-SLAPP* equals one if a firm is headquarter in a state that adopted the Anti-SLAPP law in a year. *State environmental NGOs* is the number of environmental activist NGOs in the headquarter state in a year. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Market leverage* is the ratio of long-term and short-term debt to the sum of long-term debt, short-term debt, and market value of equity. Variable definitions are in Appendix A. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 6. The moderating effects of environmental performances

Dependent variable	Market leverage			
	(1)	(2)	(3)	(4)
NGO environmental activism × Environmental overall rating	0.054** (0.023)			
Environmental overall rating	-0.006 (0.011)			
NGO environmental activism × Environmental emission rating		0.043** (0.019)		
Environmental emission rating		-0.003 (0.011)		
NGO environmental activism × Environmental resource use rating			0.045** (0.020)	
Environmental resource use rating			-0.016 (0.012)	
NGO environmental activism × Environmental innovation rating				0.025 (0.016)
Environmental innovation rating				-0.012 (0.010)
NGO environmental activism	-0.047** (0.018)	-0.035** (0.014)	-0.034** (0.014)	-0.019* (0.010)
Controls	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y
Adjusted R ²	0.853	0.853	0.853	0.853
Observations	11,127	11,127	11,127	11,127

Note: This table presents the results of estimating the effect of NGO environmental activism on financial leverage with environmental performance. The sample consists of firms from the intersection of SIGWATCH data, Refinitiv data, and Compustat data for the period 2009-2023. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Market leverage* is the ratio of long-term and short-term debt to the sum of long-term debt, short-term debt, and market value of equity. *Environmental overall rating* is Refinitiv's composite rating of the firm's overall environmental performance. *Environmental emission rating* is Refinitiv's rating of emissions management. *Environmental resource use rating* is Refinitiv's rating of resource use efficiency. *Environmental innovation rating* is Refinitiv's rating of environmental technology innovation. Variable definitions are in Appendix C. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 7. Information revelation of NGO activism on environmental liabilities

Panel A: The NGO effect on environmental incident exposure	
Dependent variable	Exposed environmental incidents (1)
NGO environmental activism	0.640*** (0.074)
Controls	Y
Firm FE	Y
Industry $\times$ Year FE	Y
Adjusted R ²	0.695
Observations	40,974

Panel B: The effect of environmental incident on leverage	
Dependent variable	Market leverage (1)
Exposed environmental incidents	-0.004*** (0.001)
Controls	Y
Firm FE	Y
Industry $\times$ Year FE	Y
Adjusted R ²	0.791
Observations	40,974

Note: Panel A presents the results of estimating the effect of NGO environmental activism on the exposure of environmental incidents. Panel B presents the results of estimating the effect of environmental liabilities on leverage. The sample consists of firms from the intersection of SIGWATCH data, RepRisk data, and Compustat data for the period 2009-2023. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Exposed environmental incidents* is the number of environmental controversies and allegations. *Market leverage* is the ratio of long-term and short-term debt to the sum of long-term debt, short-term debt, and market value of equity. Variable definitions are in Appendix C. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 8. The effect of NGO environmental activism on credit ratings

Dependent variable	Credit rating (1)
NGO environmental activism	-0.176** (0.089)
Controls	Y
Firm FE	Y
Industry $\times$ Year FE	Y
Adjusted R ²	0.957
Observations	6,490

Note: This table presents the results of estimating the effect of NGO environmental activism on S&P long-term issuer credit ratings. The sample consists of firms from the intersection of SIGWATCH data and Compustat data for the period 2009-2023. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Credit rating* is the numerical rank of S&P long-term issuer credit ratings, ranging from 1 (lowest credit quality, default) to 20 (highest credit quality, AAA). Variable definitions are in Appendix C. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 9. The effect of NGO environmental activism on cost of debt

Dependent variable	Loan spread	
	(1)	(2)
NGO environmental activism	18.235** (8.446)	16.487** (7.648)
Controls	Y	Y
Loan type FE	Y	Y
Loan purpose FE	N	Y
Firm FE	Y	Y
Industry $\times$ Year FE	Y	Y
Adjusted R ²	0.652	0.687
Observations	7,690	7,690

Note: This table presents the results of estimating the effect of NGO environmental activism on loan spread. The sample consists of firms from the intersection of SIGWATCH data, DealScan data, and Compustat data for the period 2009-2023. *NGO environmental activism* equals 1 for the firm-year observations within the following 2 years after activism, and zero otherwise. *Loan spread* is the amount the borrower pays in basis points over LIBOR for each dollar drawn down from the loan facility. Variable definitions are in Appendix C. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 10. Robustness tests

	Alternative DID models			Alternative event windows	
	Interaction Weighted Estimation	Imputation Estimation	Two- Stage Least Squares	Post 1 year	Post 3 years
Dependent variable	(1)	(2)	(3)	(4)	(5)
NGO environmental activism	-0.015*** (0.005)	-0.020*** (0.006)	-0.018*** (0.005)	-0.015** (0.006)	-0.020*** (0.007)
Controls	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Industry $\times$ Year FE	Y	Y	Y	Y	Y
Adjusted R ²	0.791	0.792	0.792	0.791	0.791
Observations	40,974	40,819	40,974	40,646	41,291

Note: This table presents the results of estimating the effect of NGO environmental activism on financial leverage with alternative DID models and alternative event windows. The sample consists of firms from the intersection of SIGWATCH data and Compustat data for the period 2009-2023. Column 1 estimates an interaction-weighted two-way fixed effects model (Sun and Abraham, 2020). Column 2 estimates a imputation-based model (Borusyak et al., 2022). Column 3 estimates a modified two-way fixed effects model with heterogeneous treatment effects (Freyaldenhoven et al., 2019). Column 4 retains the following 1 year after activism. Column 5 retains the following 3 years after activism. *Market leverage* is the ratio of long-term and short-term debt to the sum of long-term debt, short-term debt, and market value of equity. Variable definitions are in Appendix C. Standard errors are clustered at the firm level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Internet Appendix (??)

Table A1. NGO environmental activists

Top 10 NGO activists	NGO influence	Chief manager
Sierra Club U.S.A.	Global coalition (Highest)	Ben Jealous
Rainforest Action Network	Global coalition (Highest)	Ginger Cassady
Friends of the Earth U.S.A.	Global coalition (Highest)	Erich Pica
Greenpeace U.S.A.	Global coalition (Highest)	Sushma Raman
Stand.earth	Global coalition (Highest)	Todd Paglia
Natural Resources Defense Council	Global coalition (Highest)	Manish Bapna
Earthjustice	Global coalition (Highest)	Abigail Dillen
Oil Change International	Global coalition (Highest)	Elizabeth Bast
Food & Water Watch	Global coalition (Highest)	Wenonah Hauter
Environmental Working Group	Global coalition (Highest)	Ken Cook

Table A2. NGO environmental activism examples

Target company	Target year	NGO(s) involved	Issue	Campaign description
Duke Energy	2013	Southern Environmental Law Center, Cape Fear River Watch, Sierra Club, and Waterkeeper Alliance	Groundwater pollution	“Conservation groups today filed in court to join North Carolina’s complaint against Duke Energy Progress over its illegal groundwater pollution from coal ash at its Sutton facility near Wilmington, N.C. The coal waste has contaminated the area with arsenic, selenium, and other toxic pollutants, which threaten a local community’s drinking water supply.
Costco	2020	SumOfUs, Natural Resources Defense Council, Mighty Earth, Stand.earth, and Care2	Forest destruction in production and supply chain	“On Wednesday, January 22, activist organizations will unite outside Costco’s annual general shareholder meeting to challenge the leading retailer on its commitment to protecting the environment. The coalition will hold a rally and deliver over 55,000 signatures urging the retailer to stop deforestation in its own products and in its supply chain.
Exxon Mobil	2015	Various environmental groups, civil-rights organizations, and indigenous peoples’ movements	Fossil fuels, climate change	“Leaders of many of the country’s largest environmental groups, civil rights organizations, and indigenous peoples movements issued a joint letter today calling on the Department of Justice to investigate ExxonMobil, after revelations that the company knew about climate change as early as the 1970s, but chose to mislead the public about the crisis in order to maximize their profits from fossil fuels.
Chevron	2013	40+ environmental and civil-rights organizations	Environmental damage, human-rights violations	“We, the undersigned organizations, and individuals, condemn the actions by Chevron in its efforts to silence critics and ignore a \$9.5 billion judgment against it for environmental damage in the Ecuadorian Amazon. Chevron’s actions set a dangerous precedent and represent a growing and serious threat to the ability of civil society to hold corporations accountable for their misdeeds around the world. Since Chevron launched its attack on those who have been working for decades to pressure the company to clean up the environmental damage caused by its operations in the Ecuadorian Amazon, independent journalists have been forced to turn over their material and nonprofit watchdog groups have faced massive legal actions designed to cripple their ability to work and undermine their ability to grow support for their efforts.” https://amazonwatch.org/news/2013/1218-chevrons-threat-to-open-society
Whole Foods	2020	Environment America Research & Policy	Plastic waste	“Environment America Research & Policy Center, U.S. PIRG Education Fund and other nonprofits launched a national campaign on Tuesday

Center, PIRG
Education Fund, and
other NGOs

calling on Whole Foods to change its practices on plastic packaging. The groups' decision comes after the supermarket chain received an 'F' for its policies on single-use plastic packaging from As You Sow, an environmental shareholder advocacy nonprofit. Along with the Plastic Pollution Coalition, BRINGiT, and Student PIRG chapters from coast-to-coast, the organizations are specifically pressing Whole Foods to lead by eliminating single-use packaging from store shelves. The goal is to highlight the importance of this issue for consumers in all parts of the country. Some members of the coalition, along with Greenpeace, have already sent a letter to Whole Foods' CEO John Mackey urging him to take action on tackling plastic waste." <https://environmentamerica.org/media-center/groups-urge-whole-foods-put-planet-over-plastic-by-eliminating-plastic-packaging/>

Tucson Electric Power	2015	Sierra Club, Tierra Y Libertad Organization, Robert Bulechek Energy Management, Borderlands Brewing Co., Solar Possibilities Consulting, Sustainable Tucson, Casa Maria Tucson, 350.org Tucson, Sky Island Alliance, Wild Earth Guardians, and Great Old Broads for Wilderness	Divestment from the San Juan Generating Station
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"A coalition of Arizona community groups and businesses today called on Tucson Electric Power (TEP) to divest from the out-of-state San Juan Generating Station, a coal-fired power plant, and instead to commit to local, clean energy solutions like rooftop solar. Over the past few weeks, support for continued burning of coal at the San Juan Generating Station has evaporated as admissions by the Public Service Company of New Mexico's (PNM) have revealed serious financial risks for the future of the plant. Earlier this month, the home city of the plant, Farmington, New Mexico, abandoned its plans to acquire an increased portion of the plant due to reliability concerns and the huge costs that would be passed onto the community. Other New Mexico stakeholders have also pulled away from an agreement that would continue PNM's use of coal at the plant, citing the overall uncertainty about San Juan's operations. TEP owns half of one unit of the plant, which presents huge financial risks to local Tucson ratepayers if the utility continues its investment in the expensive, outdated coal-fired power plant. 'Our community faces enormous threats if TEP leads us down a path of continued reliance on coal at the San Juan Generating Station,' said Dan Millis, Tucson organizer for Sierra Club's Grand Canyon (Arizona) Chapter. 'By continuing our reliance on an expensive out-of-state coal plant, TEP is locking our community into a future of more coal and expensive rate hikes that will threaten the pocketbooks of families in our region. It's time for TEP to protect ratepayers here in Arizona by getting

out of a dirty, expensive out-of-state coal plant and investing in local clean energy such as solar.’

‘I strongly encourage TEP to divest from the San Juan plant. Coal is an energy source of the past. TEP should invest in clean, renewable energy sources that will sustain our future,’ said Rev. Lisa McDaniel-Hutchings, Co-Executive Director, Unitarian Universalist Justice Arizona Network (UUJAZ).” <https://content.sierraclub.org/press-releases/2015/02/arizona-groups-urge-tep-divest-san-juan-coal-plant>

Table A3. Validity tests: the adoption timing of anti-SLAPP

Dependent variable	Number of years to adopting anti-SLAPP
	(1)
State NGO environmental activism	-2.485 (2.379)
State unemployment rate	-31.95 (22.65)
Democratic state	1.846** (0.758)
Business combination law	1.207 (1.322)
Right-to-work law	1.206 (0.893)
Universal demand law	-1.303 (0.883)
Whistleblower law	0.412 (1.240)
Inevitable disclosure doctrine	0.318 (0.765)
Non-compete enforceability	2.001 (3.153)
Observations	293

Note: This table presents the results from Exponential hazard models where the “failure” event is enacting anti-SLAPP law by a state in a year. The dependent variable is the number of years to the adoption of anti-SLAPP law. The sample is at the state level, and a state is dropped from the sample once it adopted the anti-SLAPP law. Our main explanatory variable, *State NGO environmental activism*, is the number of NGO environmental activism campaigns targeting firms of a state in a year. Standard errors are clustered at the state level and are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.